Double, Triple and Half angle Identities

sine	cosine	Tangent
$\sin 2A = 2\sin A\cos A$	$\cos 2A = \cos^2 A - \sin^2 A$	$\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$
$\sin 2A = \dfrac{2\tan A}{1+\tan^2 A}$	$\cos 2A = 2\cos^2 A - 1$	$\tan 3A = \dfrac{3\tan A - \tan^3 A}{1-3\tan^2 A}$
$\sin 3A = 3\sin A - 4\sin^3 A$	$\cos 2A = 1 - 2\sin^2 A$	
	$\cos 2A = \dfrac{1-\tan^2 A}{1+\tan^2 A}$	
	$\cos 3A = 4\cos^3 A - 3\cos A$	
$\sin\theta = 2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}$	$\cos\theta = \cos^2\dfrac{\theta}{2} - \sin^2\dfrac{\theta}{2}$	$\tan\theta = \dfrac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$
$\sin\theta = \dfrac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$	$\cos\theta = 2\cos^2\dfrac{\theta}{2} - 1$	
	$\cos\theta = 1 - 2\sin^2\dfrac{\theta}{2}$	

Trigonometric Equation	General solution
$\sin\theta = \sin\alpha$, where $\alpha \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$	$\theta = n\pi + (-1)^n\alpha, n \in \mathbb{Z}$
$\cos\theta = \cos\alpha$, where $\alpha \in [0, \pi]$	$\theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$
$\tan\theta = \tan\alpha$, where $\alpha \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$	$\theta = n\pi + \alpha, n \in \mathbb{Z}$

Law of sine	Law of cosine	Law of Tangent
$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = 2R.$	$\cos A = \dfrac{b^2+c^2-a^2}{2bc};$	$\tan\dfrac{A-B}{2} = \dfrac{a-b}{a+b}\cot\dfrac{C}{2}$
	$\cos B = \dfrac{a^2+c^2-b^2}{2ac};$	$\tan\dfrac{B-C}{2} = \dfrac{b-c}{b+c}\cot\dfrac{A}{2}$
	$\cos C = \dfrac{a^2+b^2-c^2}{2ab}$	$\tan\dfrac{C-A}{2} = \dfrac{c-a}{c+a}\cot\dfrac{B}{2}$

In a $\triangle ABC$, we have

(i) $a = b\cos C + c\cos B$, $b = c\cos A + a\cos C$ and $c = a\cos B + b\cos A$ [**Projection formula**].

(ii) $\triangle = \dfrac{1}{2}ab\sin C = \dfrac{1}{2}bc\sin A = \dfrac{1}{2}ac\sin B$ [**Area of the triangle**].

(iii) $\triangle = \sqrt{s(s-a)(s-b)(s-c)}$

(iv) $\sin\dfrac{A}{2} = \sqrt{\dfrac{(s-b)(s-c)}{bc}}$, $\cos\dfrac{A}{2} = \sqrt{\dfrac{s(s-a)}{bc}}$, $\tan\dfrac{A}{2} = \sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}}$, where s is the semi-perimeter of $\triangle ABC$ given by $s = \dfrac{a+b+c}{2}$. [**Half Angle formula**].